

Sparsh Kumar Gupta

Instrumentation Engineer passionate about tech across **hardware, software, and product design**, with hands-on experience in UI/UX, Building Automation Systems (BAS), IoT, and Machine learning projects.

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Education

Techno Main Salt Lake (Kolkata, India) — 2022-2026

University: Maulana Abul Kalam Azad University of Technology, West Bengal

Bachelor of Technology | CGPA - 8.13 | Electronics and Instrumentation Engineering (EIE)

Work Experience

Johnson Controls — Oct 2026 - Present

Graduate Engineer Trainee - Global Complex Engineering - Post Sales

Johnson Controls — Mar 2026 - Apr 2026

Controls Domain | Internship

- Worked on Building Automation Systems (BAS), creating hardware submittals from mechanical drawings and specifications.
- Configured HVAC systems including AHUs, VAVs, chillers, and boilers using Solution Navigator.
- Developed wiring diagrams, system layouts, and schedules using Visio and Excel.
- Built Metasys UI graphics and gained hands-on experience with SCT and CCT through training projects.

Scalenow Technosolutions Pvt. Ltd. (Hybrid – India) — Nov 2023 - Oct 2025

UI/UX Designer | Internship

- Designed user flows and high-fidelity screens for a travel-tech marketplace connecting operators and travelers.
- Designed the My Councilor app, enabling ward councilors to efficiently manage and resolve ward-related issues in real time.
- Created a visitor management system aimed at improving residential society security and simplifying entry processes.
- Led multiple website design projects, delivering interfaces aligned with brand identity and client objectives.
- Conducted user research and designed a quotation making system aimed at improving sales team efficiency .

My SupportRoom Private Limited (Remote – United Kingdom) — Mar 2023 – May 2024

UI/UX Designer | Internship

- Led UX research and UI design for SupportFit, a personalized health and fitness tracking app.
- Designed and launched SupportFit v2 with improved usability and user-specific health features.
- Developed a personalized resource center to improve user engagement and self-service.
- Designed interfaces for chatrooms, video calls, voice notes, calendar sync, and web integration, improving the overall product.

Projects

- Developed a computer vision–based ML model with macOS, Windows, and Android applications to detect and classify six types of plastics: HDPE, LDPE, PET, PVC, PP, and PS.
- Made a smart dustbin with automated lid opening via proximity detection and real-time fill level monitoring, integrated with the Blynk IoT app using ESP8266, HC-SR04 sensor, and servo motor.

Trainings and Awards

IEEE Research Paper — Feb 2026

Project Name: RecycleNet

Presented a research paper on a computer vision–based plastic classification system at ICCECE 2026; paper accepted for publication in the IEEE Digital Library.

Mitsubishi Electric Factory Automation — Nov 2025

Completed certified training in Mitsubishi Electric Factory Automation products, including FX/Q PLCs, GOT2000 & GS Series HMIs, and D700/E700 AC Drives. Gained hands-on experience in PLC programming, HMI development, VFD configuration, and industrial automation applications.

Hult Prize Competition | Winner — Feb 2023

Project Name: “Fibre” | Old Clothes – New Possibilities

Campus Level Winner

Regional Level @ Milan, Italy

UI / UX Design Competition | Winner — Mar 2023

Event : The Next Billion | Theme: Redesigning a Famous Application

Enhanced the user experience by redesigning the user interface of Amazon.

Martial Arts (Kyodokyokushin Karate)

Maintained a consistent training regimen, building physical strength, mental resilience, and determination.

International Level – 1st Runners Up

State Level – 1st Runners Up

Skills

Technical Skills - Building Automation Systems (BAS) HVAC Controls Machine Learning IoT Prototyping C (Basic)

Design & Product - UI/UX Design User Research Wireframing Prototyping Design Systems

Tools & Software - Figma Framer Solution Navigator Metasys UI SCT CCT Visio PLC HMI

Frontend - HTML CSS